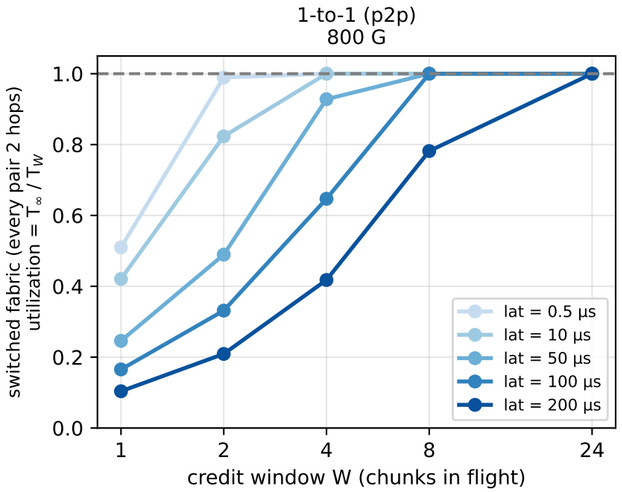
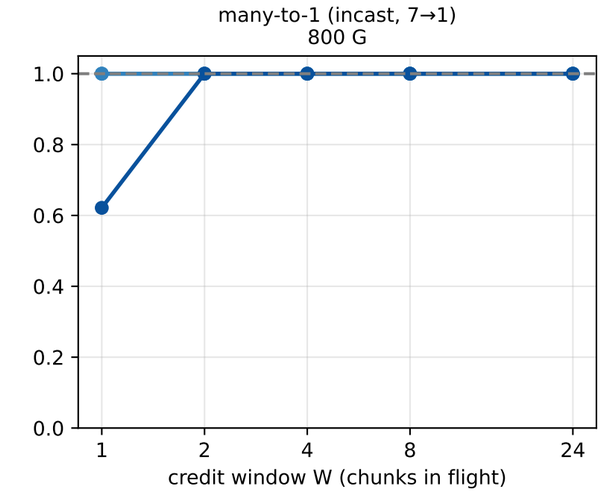
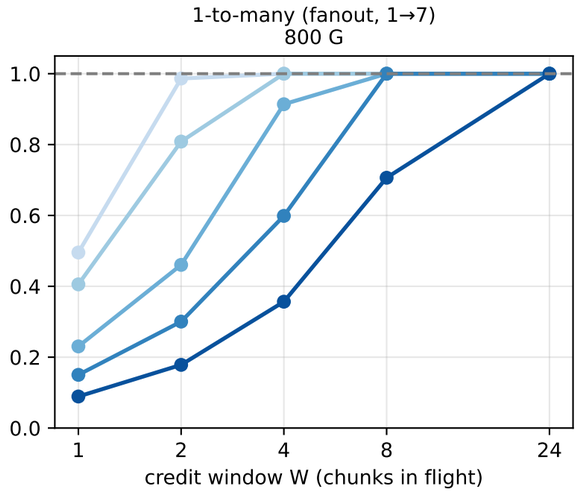
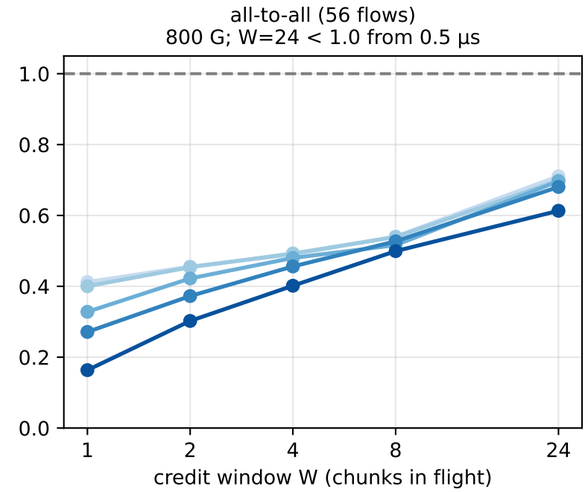
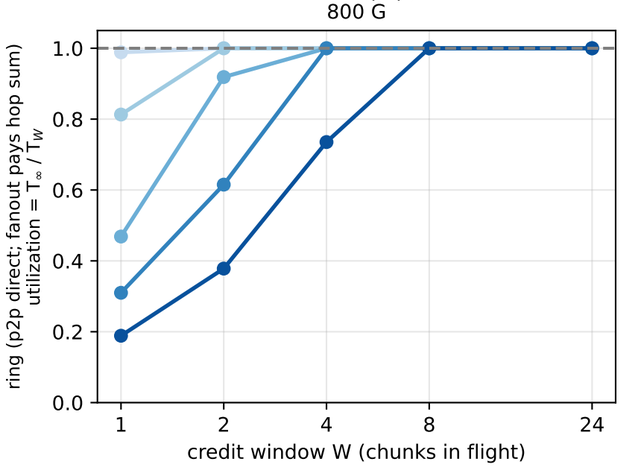
Credit flow control across traffic patterns at FIXED bandwidth (800 G, 8-GPU Su fabric, 256 MiB per flow):
utilization vs credit window W and one-way latency — dedicated (p2p/incast) vs shared (fanout/alltoall) windows, ring vs switched

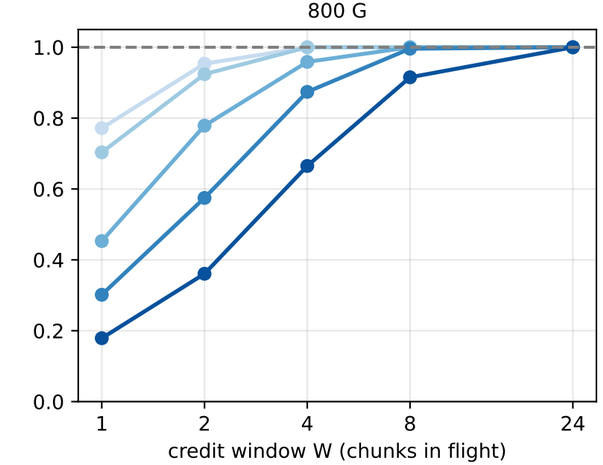


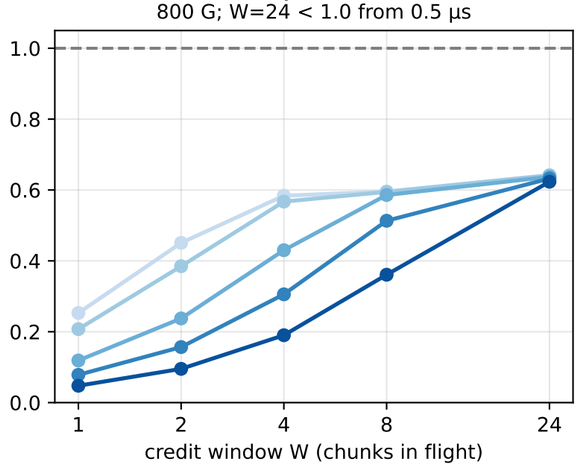


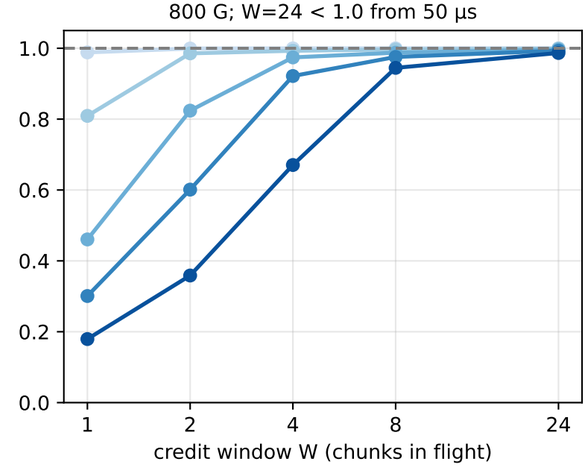


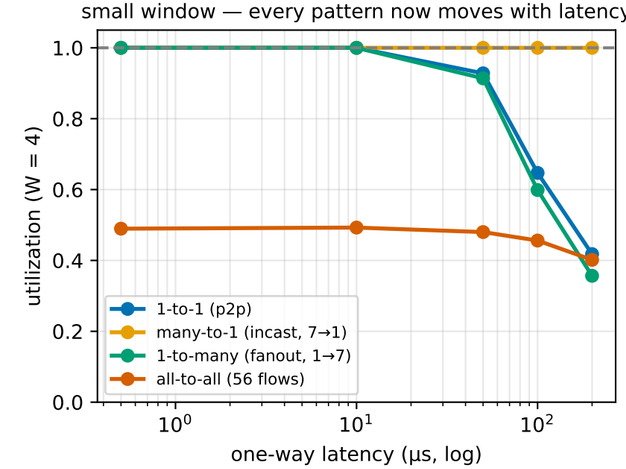


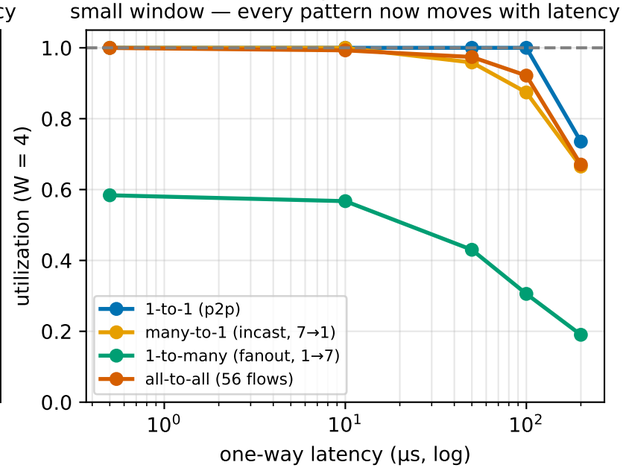












Utilization at 800 G, W = 4

lat (µs):	0.5	10	50	100	200
--- switched ---					
1-to-1 (p2p)	1.000	1.000	0.928	0.647	0.418
many-to-1 (incast, 7→1)	1.000	1.000	1.000	1.000	1.000
1-to-many (fanout, 1→7)	1.000	1.000	0.914	0.599	0.357
all-to-all (56 flows)	0.490	0.493	0.480	0.456	0.402
--- ring ---					
1-to-1 (p2p)	1.000	1.000	1.000	1.000	0.735
many-to-1 (incast, 7→1)	1.000	1.000	0.959	0.874	0.665
1-to-many (fanout, 1→7)	0.584	0.567	0.430	0.306	0.190
all-to-all (56 flows)	1.000	0.993	0.974	0.922	0.671

Reading rows 1-2: the W needed for full utilization shifts right ~linearly with latency (credit RTT = hops·(ser+2·lat)); fanout/alltoall need ~W×flow-count credits (shared 1-VC window) — the window, not the link, is the bottleneck; at W=24 only alltoall (switched) / fanout (ring) remain capped (CEIL_W=24 for that ceiling view).

